Part 1: Basic iptables usage

```
user@ubuntu-atker:~$ sudo iptables -L
[sudo] password for user:
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

```
user@ubuntu-atker:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere             ctstate RELATED,ESTABLI
SHED
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:domain
ACCEPT     udp  --  anywhere             anywhere             udp dpt:domain
ACCEPT     icmp --  anywhere             anywhere             ctstate NEW,RELATED,EST
ABLISHED icmp destination-unreachable
ACCEPT     icmp --  anywhere             anywhere             ctstate NEW,RELATED,EST
ABLISHED icmp time-exceeded
ACCEPT     icmp --  anywhere             anywhere             ctstate NEW,RELATED,EST
ABLISHED icmp parameter-problem
DROP       all  --  anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

```
user@ubuntu-atker:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
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ACCEPT     all  --  anywhere             anywhere
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ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:domain
ACCEPT     udp  --  anywhere             anywhere             udp dpt:domain
ACCEPT     icmp --  anywhere             anywhere             ctstate NEW,RELATED,EST
ABLISHED icmp destination-unreachable
ACCEPT     icmp --  anywhere             anywhere             ctstate NEW,RELATED,EST
ABLISHED icmp time-exceeded
ACCEPT     icmp --  anywhere             anywhere             ctstate NEW,RELATED,EST
ABLISHED icmp parameter-problem
DROP       all  --  anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

```
user@ubuntu-atker:~$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 1 packets, 73 bytes)
 pkts bytes target     prot opt in     out     source               destination

    1    73 ACCEPT     all  --  lo     any     anywhere             anywhere

    0     0 ACCEPT     all  --  any    any     anywhere             anywhere
    ctstate RELATED,ESTABLISHED
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere
    tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere
    tcp dpt:domain
    0     0 ACCEPT     udp  --  any    any     anywhere             anywhere
    udp dpt:domain
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
    ctstate NEW,RELATED,ESTABLISHED icmp destination-unreachable
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
    ctstate NEW,RELATED,ESTABLISHED icmp time-exceeded
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
    ctstate NEW,RELATED,ESTABLISHED icmp parameter-problem
    0     0 DROP       all  --  any    any     anywhere             anywhere


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

The rules are still active after installing the package and rebooting

Part 2

```
user@ubuntu-atker:~$ sudo iptables -t mangle -L -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

user@ubuntu-atker:~$ sudo iptables -Z
user@ubuntu-atker:~$ sudo iptables -t mangle -Z
```

9. Nmap scan

```
user@Ubuntu-client1:~$ sudo nmap -sN 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-27 13:01 UTC
Nmap scan report for 192.168.56.103
Host is up (0.00067s latency).
Not shown: 999 open|filtered tcp ports (no-response)
PORT   STATE  SERVICE
22/tcp closed ssh
MAC Address: 08:00:27:16:6A:B8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.93 seconds
user@Ubuntu-client1:~$ sudo nmap -sW 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-27 13:01 UTC
Nmap scan report for 192.168.56.103
Host is up (0.00059s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE  SERVICE
22/tcp closed ssh
53/tcp closed domain
MAC Address: 08:00:27:16:6A:B8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 5.02 seconds
```

4k packet dropped

```
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

   24  2665 ACCEPT     all  --  lo     any     anywhere             anywhere

   56 17985 ACCEPT     all  --  any    any     anywhere             anywhere
  ctstate RELATED,ESTABLISHED
    4   160 ACCEPT     tcp  --  any    any     anywhere             anywhere
  tcp dpt:ssh
    5   200 ACCEPT     tcp  --  any    any     anywhere             anywhere
  tcp dpt:domain
    0     0 ACCEPT     udp  --  any    any     anywhere             anywhere
  udp dpt:domain
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp destination-unreachable
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp time-exceeded
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp parameter-problem
 4034  167K DROP       all  --  any    any     anywhere             anywhere


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 126 packets, 13382 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

```
user@ubuntu-atker:~$ sudo iptables -t mangle -L -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

```
user@Ubuntu-client1:~$ sudo nmap -sN 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-27 13:06 UTC
Nmap scan report for 192.168.56.103
Host is up (0.00s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)
MAC Address: 08:00:27:16:6A:B8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 21.63 seconds
```

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```
user@ubuntu-atker:~$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

    1    73 ACCEPT     all  --  lo     any     anywhere             anywhere

    4  1330 ACCEPT     all  --  any    any     anywhere             anywhere
  ctstate RELATED,ESTABLISHED
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere
  tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere
  tcp dpt:domain
    0     0 ACCEPT     udp  --  any    any     anywhere             anywhere
  udp dpt:domain
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp destination-unreachable
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp time-exceeded
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp parameter-problem
    4   290 DROP       all  --  any    any     anywhere             anywhere


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 7 packets, 773 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

```
user@ubuntu-atker:~$ sudo iptables -t mangle -L -v
Chain PREROUTING (policy ACCEPT 10 packets, 1766 bytes)
 pkts bytes target     prot opt in     out     source               destination

 2000 80000 DROP       all  --  any    any     anywhere             anywhere
  ctstate INVALID
    0     0 DROP       tcp  --  any    any     anywhere             anywhere
  tcp flags:!FIN,SYN,RST,ACK/SYN ctstate NEW

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

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```
user@Ubuntu-client1:~$ sudo nmap -sW 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-27 13:08 UTC
Nmap scan report for 192.168.56.103
Host is up (0.0010s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 08:00:27:16:6A:B8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 21.60 seconds
```

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```
user@ubuntu-atker:~$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

    1    73 ACCEPT     all  --  lo     any     anywhere             anywhere

   17  4582 ACCEPT     all  --  any    any     anywhere             anywhere
  ctstate RELATED,ESTABLISHED
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere
  tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere
  tcp dpt:domain
    0     0 ACCEPT     udp  --  any    any     anywhere             anywhere
  udp dpt:domain
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp destination-unreachable
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp time-exceeded
    0     0 ACCEPT     icmp --  any    any     anywhere             anywhere
  ctstate NEW,RELATED,ESTABLISHED icmp parameter-problem
    7   607 DROP       all  --  any    any     anywhere             anywhere


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 23 packets, 2165 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

```
user@ubuntu-atker:~$ sudo iptables -t mangle -L -v
Chain PREROUTING (policy ACCEPT 38 packets, 7723 bytes)
 pkts bytes target     prot opt in     out     source               destination

 2000 80000 DROP       all  --  any    any     anywhere             anywhere
  ctstate INVALID
 2000 80000 DROP       tcp  --  any    any     anywhere             anywhere
  tcp flags:!FIN,SYN,RST,ACK/SYN ctstate NEW

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```